

Global and local feature extraction of urban historical spatial perception using large language models: A case study of Harbin Central Street District

Haixuan Zhu^{*}, Jiang Chang, Xinyu An, Shilin Li

School of Architecture and Design, Harbin Institute of Technology, China

Key Laboratory of Cold Region Urban and Rural Human Settlement Environment Science and Technology, Ministry of Industry and Information Technology, China

ARTICLE INFO

Keywords:

Large language models (LLMs)
BERTopic
Spatial perception, global and local feature
Harbin Central Street
Harbin Stalin Park

ABSTRACT

Spatial perception—the process by which individuals interpret and interact with their spatial environment—is a cornerstone of urban studies, forming the foundation for effective urban planning and policy-making. This study introduces innovative, digital methodologies for spatial perception analysis by integrating advanced technologies, including Large Language Models (LLMs) and BERTopic modeling. Using Harbin Central Street District, a historically significant urban area celebrated for its architectural heritage and scenic landscapes, as a case study, this research develops a replicable and scalable workflow for analyzing spatial perception. The study identifies global spatial perception patterns, such as pedestrian dynamics, green spaces, and cultural landmarks, alongside local themes unique to Central Street and Stalin Park. Key contributions include leveraging LLMs to generate synthetic text data from urban imagery, applying high-dimensional semantic vector models for topic analysis, and utilizing dynamic modeling techniques to extract both global and local spatial features. The findings reveal how historical architecture, commercial vibrancy, and natural environments interact to shape urban experiences, aligning closely with established spatial metrics. This research provides actionable insights for urban planning and policy, emphasizing heritage preservation, pedestrian-oriented design, and adaptive strategies. By bridging AI-driven technologies with urban research, this study offers a scalable, data-driven framework for analyzing spatial perceptions, advancing the digital transformation of cities, and promoting sustainable, inclusive urban development in the AI era.

1. Introduction

Spatial perception refers to the processes through which individuals perceive, interpret, and interact with their spatial environments. It serves as a cornerstone for creating livable urban spaces and forms the theoretical foundation for sound urban policies, making it a critical focus in urban studies (Gehl, 1987; Gibson, 1950; Lynch, 1960; Tuan, 1977). With rapid advancements in artificial intelligence (AI) and computational technologies, urban planning is entering an era of digital transformation and algorithmic innovation. As cities act as spatial receptacles for social, economic, and cultural activities (De Jong et al., 2024), this evolution necessitates a shift in paradigms, theories, and methodologies for studying spatial perception to meet the demands of AI-driven urban governance.

The emergence of large language models (LLMs), such as ChatGPT, has further accelerated this transformation. As Reid Hoffman aptly described, “AI is the cognitive industrial revolution—the steam engine

of the mind” (Sperling, 2024). Multimodal large language models (MLLMs) like GPT-4o exhibit unprecedented capabilities in semantic understanding and spatial perception, particularly in processing visual data—a breakthrough likened to the evolutionary development of vision in biological systems (Li, 2024). These advancements have opened new horizons for urban research, enabling a redefinition of spatial perception theories and methodologies rooted in AI and computer science, with profound implications for urban planning and policymaking (Ito et al., 2024).

Human-centric urban spaces are primarily experienced at meso- and micro-scales, which are the main dimensions of daily spatial perception. Historical urban districts, as archetypes of such spaces, offer rich cultural heritage and high-quality spatial environments, making them stable and representative public spaces. Against this backdrop, this study focuses on Harbin Central Street District as a case study to explore the innovative application of LLMs and pre-trained semantic models in spatial perception research.

^{*} Corresponding author at: NO. 92, Xidazhi Street, Nangang District, Harbin, Heilongjiang Province, China.

E-mail address: zhuhx1979@hit.edu.cn (H. Zhu).

The primary objectives of this study are as follows: First, Synthetic Text Data Generation: Develop a methodology to transform spatial images into descriptive text using GPT-4o's robust image-to-text capabilities. Second, Representation and Measurement of Spatial Perception: Use BERTopic for semantic modeling and c-TF-IDF to quantitatively measure spatial perception themes. Third, Extraction of Global and Local Themes: Apply dynamic topic modeling techniques to extract global themes representing overarching spatial patterns and local themes capturing context-specific nuances. Fourth, Case Study and Workflow Development: Conduct an empirical study of Harbin Central Street District to establish a representative case study and construct a replicable, scalable workflow for spatial perception research.

This research aims to pioneer new paradigms for spatial perception in the AI era and develop a comprehensive digital workflow for spatial perception analysis. The findings are expected to provide theoretical insights for designing high-quality urban spaces and data-driven guidance for informed urban planning and policy decision-making.

2. Literature review

Research on spatial perception traces back to Gibson's foundational work, *The Perception of the Visual World* (1950), which laid the groundwork for studying visual perception. Subsequent publications, such as *The Ecological Approach to Visual Perception*, established a sensory-focused framework that remains central to the field. In urban planning and policy research, spatial perception—defined as individuals' synesthetic experiences of urban spaces (Garling, 1972; Lynch, 1960; Tuan, 1977)—has consistently been a dynamic and critical area of inquiry.

This review examines spatial perception research through three core dimensions: data sources, methodologies, and modes of representation, with an emphasis on advancements driven by digital innovation.

2.1. Statistical analysis based on survey data

Early research on spatial perception relied on traditional data collection methods, including field surveys, face-to-face interviews (structured and semi-structured), questionnaires, and telephone interviews (Cresswell, 1992; McGinn et al., 2007; Montello et al., 2003). These approaches provided foundational insights, capturing individual experiences and interpreting the subtle nuances of spatial environments. Survey-based studies generated either unstructured textual data (Eliasson et al., 2007) or quantified perceptions using tools like N-point Likert scales, which measured responses on a continuum from "positive" to "negative" (Lenzholzer & Koh, 2010). Analytical methods such as ANOVA, descriptive statistics, reliability analysis, hypothesis testing, and multivariate analysis were widely applied to these datasets (Böcker et al., 2015). While these methods established the groundwork for spatial perception research, they were constrained by their time-intensive nature, high costs, and small sample sizes, limiting their applicability to localized studies.

2.2. Cognitive mapping and spatial representation

Lynch's (1960) *The Image of the City* introduced mental or cognitive maps as a tool to represent how individuals cognitively understand and interpret space. Unlike geometric or geographic descriptions, cognitive maps reflect subjective meanings assigned to environments, which are often challenging to quantify. Lynch's work inspired the development of various mental and perception mapping techniques (Downs & Stea, 1973; Heft, 2013; Kaplan, 1973; Kitchin & Blades, 2002).

In perception mapping studies, respondents were often asked to mark areas of significance on maps. When participants faced difficulty drawing directly, researchers annotated maps based on verbal descriptions provided by respondents (Lenzholzer & Koh, 2010). To spatially contextualize these perceptions, outcomes were correlated

with GPS data collected during interviews. By integrating textual and spatial data, researchers generated GIS-based perception maps, offering valuable insights into how individuals assign meaning to various spatial typologies (Klemm et al., 2015).

2.3. Machine learning-based big data analysis

The rapid advancements in computational techniques and the availability of diverse behavioral and spatial datasets (e.g., trajectory data, text, images, and location-based data) have enabled the quantification and measurement of spatial perceptions on an unprecedented scale (Ma et al., 2021). Machine learning has become a cornerstone of this transformation, enabling large-scale, data-driven explorations of human-environment interactions (Biljecki & Ito, 2021; Gonzalez et al., 2020; Nagata et al., 2020).

2.3.1. Spatial perception analysis using street view data

Street view imagery (SVI) has emerged as a vital resource for spatial perception analysis. For instance, the MIT Media Lab's Place Pulse project assessed urban environments using six emotional dimensions—beautiful, boring, depressing, lively, wealthy, and safe—providing a framework for understanding psychological responses to urban spaces (Dubey et al., 2016).

Deep learning algorithms, such as FCN, ResNet, and SegNet, process visual data through convolutional neural networks (CNNs) to identify urban features, including roads, buildings, sidewalks, trees, and other elements (Wang et al., 2019; Wu et al., 2019). These techniques establish a technical foundation for understanding how visual elements influence spatial perception. For example, Wang et al. (2022) developed a scoring model based on street imagery to quantify six emotional dimensions. These scores were further categorized into positive and negative spatial perceptions, which were integrated with spatial accessibility data to prioritize urban renewal interventions and inform planning decisions.

2.3.2. Spatial perception analysis using natural language processing

Natural Language Processing (NLP) encompasses computational methods for analyzing and representing human language and has been increasingly applied to urban studies using digital text data, such as social media posts and online reviews (Chowdhary, 2020; Liddy, 2001).

NLP-based spatial perception analysis can be categorized into two approaches: First, Topic Classification: This method categorizes spatial perceptions into predefined categories, similar to N-point Likert scales. Lexicon-based approaches identify positive or negative sentiment using sentiment dictionaries (Liu, 2012). Machine learning models, trained on annotated text datasets, enhance classification accuracy, while hybrid methods combine lexicon-based and machine learning approaches for greater flexibility and precision (Wankhade et al., 2022). Second, Semantic Analysis: Semantic analysis derives deeper meanings by grouping words and sentences based on their syntactic structures (Khurana et al., 2023; Salloum et al., 2020). Techniques such as Word2Vec, Latent Semantic Analysis (LSA), and Latent Dirichlet Allocation (LDA) identify latent themes and relationships within text data (Blei et al., 2003; Mikolov et al., 2013). Unlike Word2Vec, which focuses on individual word associations, LSA and LDA use entire document contexts, enabling a more comprehensive understanding of spatial perceptions across text datasets.

2.4. Opportunities and challenges: advancing spatial perception with MLLMs

The emergence of MLLMs, such as GPT-4o, represents a groundbreaking advancement in spatial perception research. These models integrate vision and language processing, enabling highly accurate image-to-text conversion and advanced semantic analysis, fundamentally transforming how spatial perceptions are studied (Chen et al.,

2024).

Recent developments highlight the potential of MLLMs for spatial perception analysis. For example, Malekzadeh et al. (2025) used GPT-4 to analyze over 1800 Google Street View images of Helsinki, Finland, evaluating urban attractiveness through image feature extraction and semantic segmentation. The strong correlation between GPT-4's analyses and human evaluations demonstrates its capability to capture spatial perceptions at scale with unprecedented precision and efficiency.

Building on these advancements, this study introduces three pivotal innovations:

- 1) **Synthetic Data Generation:** Leveraging GPT-4o's image-to-text capabilities to produce synthetic text data, bridging the gap between visual data and perceptual insights.
- 2) **Enhanced Topic Modeling:** Applying BERTopic for high-dimensional semantic modeling to uncover coherent and contextually rich spatial perception themes.
- 3) **Dynamic Perception Analysis:** Utilizing dynamic modeling techniques to extract and analyze global and local spatial perception topics, offering nuanced insights into spatial dynamics.

These innovations form the foundation of a replicable and scalable workflow for spatial perception research, offering a transformative approach to urban planning and policy development. By integrating cutting-edge AI technologies, this research provides a robust, data-driven framework for understanding urban environments, facilitating the creation of adaptive, inclusive, and sustainable urban strategies in the AI era.

3. Theoretical foundations

The application of MLLMs, such as GPT-4o, to spatial perception research is grounded in the principle of human-machine alignment. This principle emphasizes calibrating AI systems to align their outputs with human intentions and objectives (Gabriel, 2020). As a foundational tenet in MLLMs development, it ensures that the generated outputs meet human expectations. Models like GPT-4o exhibit exceptional capabilities in visual and spatial comprehension, producing synthetic text data that closely mirrors human perceptions of images and environments. These advanced capabilities enable GPT-4o to generate precise descriptions of spatial experiences and conduct systematic semantic analyses of human perceptions, facilitating the digital and algorithmic transformation of spatial perception research.

3.1. Semantic alignment between images and text in MLLMs

A cornerstone of utilizing MLLMs like GPT-4o in spatial perception research is their ability to semantically align images and text. Deep neural networks, particularly those used in MLLMs, increasingly converge toward shared semantic structures across different modalities. This ensures that text generated by image-to-text systems, such as GPT-4o, effectively captures spatial perceptions.

The Platonic Representation Hypothesis (Huh et al., 2024) posits that neural networks trained across modalities converge on a unified statistical representation of reality. MLLMs leverage integrated architectures to process both visual and textual data, bypassing the need for separate systems for each modality. This unified approach allows these models to construct holistic representations of the observable world, capturing the joint distribution of events that shape reality. This hypothesis aligns with the philosophical framework of convergent realism, which holds that scientific theories increasingly approximate the truth (Doppelt, 2007; Hardin & Rosenberg, 1982; Newton-Smith, 1981; Putnam, 1982).

Empirical validation of this hypothesis is provided by studies such as Malekzadeh et al. (2025), which demonstrated that GPT-4's spatial perceptions derived from Google Street View images closely align with human evaluations. These findings underscore the theoretical

foundation that MLLMs, through their unified architectures, enable accurate semantic alignment between images and text, effectively modeling spatial perceptions.

3.2. Empathy of large language models

The concept of empathy in LLMs refers to their ability to exhibit cognitive, emotional, and perspective-taking traits during interactions in social or spatial contexts (Kosinski, 2024; Xi et al., 2023). This capacity enhances their relevance in spatial perception research by providing deeper insights into human experiences, cultural contexts, and intentions.

First, Cognitive Abilities: LLMs possess advanced reasoning, problem-solving, and comprehension skills. These capabilities enable them to analyze complex urban environments, including architectural styles, historical contexts, and cultural significance. By synthesizing spatial and cultural data, LLMs uncover nuanced characteristics of spatial perception, deepening our understanding of how urban spaces are perceived.

Second, Emotional Intelligence: LLMs can interpret and simulate emotional responses based on contextual cues. This is particularly valuable in spatial perception research, where subjective emotions and feelings significantly influence spatial experiences. By generating empathetic narratives, LLMs provide richer insights into the sensory and emotional dimensions of urban spaces.

Third, Perspective Representation: LLMs can simulate the thought patterns, emotions, and behaviors of diverse stakeholders, including residents, tourists, urban planners, and historians. By adopting these varied perspectives, LLMs reveal how different groups interact with and perceive urban spaces, offering a multi-dimensional understanding of spatial perception.

Through their cognitive, emotional, and perspective-simulating capabilities, LLMs demonstrate an empathetic alignment with human perceptions. This alignment ensures that synthetic text generated by LLMs-based image-to-text technologies authentically reflects spatial perceptions, enabling the extraction of deeper insights and patterns from complex datasets.

4. Study setting

4.1. Spatial setting

Harbin, the capital of Heilongjiang Province, China, is a modern city whose growth and development were catalyzed by the construction of the Chinese Eastern Railway in 1898. By the 1920s, Harbin had evolved into a prominent political, economic, and cultural hub in Northeast Asia, serving as the administrative center for the railway. Its strategic location at the intersection of the railway's mainline (Manzhouli to Suifenhe) and its southern branch (Harbin to Lushun) propelled its rapid expansion.

Internationally, Harbin housed consulates from numerous countries, facilitated active trade with Europe and Southeast Asia, and became a cultural confluence of Chinese and Western influences. This integration transformed Harbin into a testing ground for modern urban civilization. Today, the city retains its rich historical legacy, exemplified by its Garden City designs, Art Nouveau architecture, early modernist structures, Chinese Baroque styles, and European-style churches, making Harbin an archetypal historical city and a significant cultural landmark.

The Harbin Central Street District is a celebrated historical and cultural hub, renowned for its unique blend of cultural heritage and scenic beauty (Fig. 1) (Zhu et al., 2024). The district encompasses key tourist destinations such as Saint Sophia Cathedral, Zhaolin Park, the Jewish Synagogue, the Binzhou Railway Bridge, and Sun Island Park. This study focuses on two distinct areas within the district: Central Street and Stalin Park, each offering unique characteristics for exploring spatial perception patterns.

Central Street, established in 1898, stretches 1450 m from Jingwei

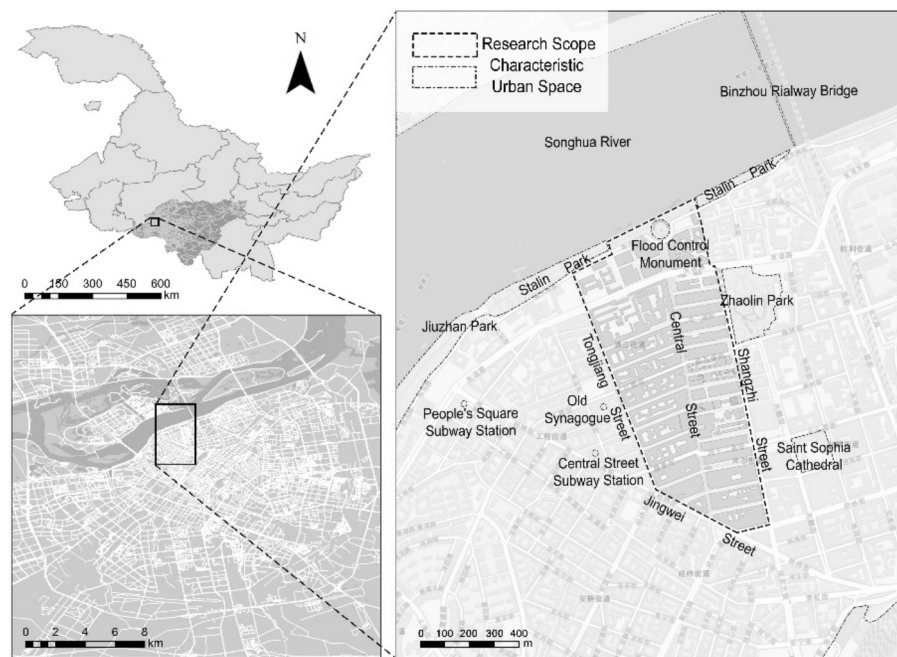


Fig. 1. Harbin Central Street District.

Street in the south to the Flood Control Monument in the north. The street, paved with granite cobblestones locally referred to as “breadstones,” is flanked by 71 European-style historical buildings. These buildings showcase a variety of architectural styles, including Renaissance, Baroque, Eclecticism, and Art Nouveau, creating a distinctly European ambiance. Converted into a pedestrian commercial street in 1996, Central Street has become a model for urban pedestrian spaces. Its vibrant mix of retail, tourism, and cultural attractions, coupled with its identity as an open-air museum of architectural art, makes it a dynamic destination for both locals and tourists.

Stalin Park, located along the Songhua River near the Flood Control Monument and facing Sun Island, is a significant greenbelt and leisure area. Established in 1953 as “Riverside Park” to commemorate Harbin’s recovery from catastrophic floods, the site previously served as a railway line and shipping terminal. The park spans 1750 m in length and 50 m in width, covering a total area of 105,000 square meters. It features Russian-style historic buildings, such as the Jiangshang Club and Riverside Restaurant, along with artistic sculptures, flowerbeds, lawns, and recreational areas. Today, Stalin Park is a popular destination for both residents and tourists, offering a harmonious blend of scenic beauty and cultural heritage.

4.2. Conceptual setting

Dynamic Topic Models (DTMs) are probabilistic time-series models designed to analyze the evolution of topics over time within large document collections (Blei & Lafferty, 2006). Unlike static models, DTMs capture temporal and spatial dynamics, making them particularly effective for identifying both global topics (derived from the entire dataset) and local topics (specific to temporal or regional subsets). This capability makes DTMs especially suitable for examining spatial perception patterns in historical urban districts.

This study employs BERTopic to model both global and local topics by utilizing c-TF-IDF scores (Grootendorst, 2022). The approach consists of two primary steps: First, Global Topic Modeling: The entire corpus is analyzed to identify global topics, which are assumed to be independent of temporal or spatial influences. This provides a comprehensive understanding of overarching themes. Documents with similar thematic content, despite contextual variations, are grouped under the same

global topic, creating a holistic view of spatial perceptions. Second, Local Topic Modeling: Building on the global topic model, local topics are identified by adjusting the term frequencies of documents within specific temporal or spatial subsets using precomputed global IDF values. This dynamic approach captures localized themes that reflect context-specific nuances while maintaining their connection to the overarching global patterns. By integrating global and local topic modeling, this dynamic methodology enables a nuanced analysis of spatial perceptions, uncovering both broad trends and localized variations. This provides a robust theoretical and algorithmic foundation for spatial perception research in complex urban environments.

5. Methodology

5.1. Data collection

The availability and reliability of data are crucial for ensuring the quality and validity of research. This study employed on-site data collection with volunteer participation for two key reasons: First, Absence of comprehensive Street View Imagery (SVI): The study sites—Central Street and Stalin Park—are pedestrian-only zones, which limits the coverage of open-source SVI platforms. For instance, Google Maps provides only partial street view images for the area, such as the segment between West 7th Street and Youyi Road. Second, Limitations of SVI in capturing real-life spatial experiences: SVI data often fail to accurately represent the immersive and dynamic experiences of individuals physically present in a space. Due to its static nature and restricted temporal scope, SVI cannot fully capture the sensory and interactive dimensions of urban environments (Malekzadeh et al., 2025).

To address these limitations, the study conducted primary data collection using a team of 10 volunteers affiliated with the authors’ institutions. The team included 2 faculty members and 8 students from the School of Architecture and Design. Data collection was carried out on July 22, 2024, at 10:00 AM under clear weather conditions. Primary data accounted for 90 % of the dataset due to its specificity and diversity, enabling a more comprehensive exploration of spatial perceptions (Ito et al., 2024).

The date and time were carefully chosen based on an analysis of

Central Street's daily activity patterns to optimize data capture. Conducting the collection on a Saturday ensured a diverse mix of local residents and tourists, while the 10:00 AM time slot avoided peak pedestrian traffic, providing clear visibility of architectural details, urban features, and pedestrian interactions.

Volunteers were provided with a concise guidance document outlining the study's objectives, methodology, and data collection protocols. The "Natural Experience" approach was employed, allowing participants to explore the space freely without predefined constraints (Dunning, 2012). Participants independently selected their starting points, routes, and areas of interest. Using smartphones, they captured images of buildings, landscapes, people, events, and commercial spaces. After data collection, participants performed initial cleaning and coding of the images to ensure alignment with the study's requirements.

5.2. Preprocessing

5.2.1. Data classification and cleaning

Following data collection, the first step involved classifying and cleaning the dataset. The collected images were organized into 10 sample sets, labeled Sam_1 through Sam_10, corresponding to each volunteer. Blurry images, duplicates, and irrelevant content were removed, resulting in 628 valid images. The dataset was then categorized by spatial distribution. For dynamic modeling, each sample set was further divided into eight spatial segments (Table 1), ensuring comprehensive global topic coverage and a clear progression of local topics.

5.2.2. Image-to-text using GPT-4o with prompt engineering

The advanced semantic understanding and spatial analysis capabilities of GPT-4o served as the foundation for generating textual data from the collected images. Since GPT-4o is a general-purpose language model, precise and detailed prompt engineering was essential to ensure high-quality outputs (Fig. 2). The process involved two stages:

First, Contextual prompt engineering: This stage involved equipping GPT-4o with domain-specific knowledge, contextual information, and task instructions (Table 2). Domain knowledge: Provided information on spatial perception, spatial scales, and the significance of urban planning to help GPT-4o generate specialized and accurate descriptions. Contextual information: Informed GPT-4o of the spatial and cultural characteristics of Central Street and Stalin Park to enhance relevance and precision. Task instructions: Ensured GPT-4o understood the study's objectives and requirements, enabling tailored and precise text generation.

Second, Few-Shot prompt engineering: To optimize GPT-4o's performance, a few-shot learning approach was adopted (Touvron et al., 2023): Zero-shot image-to-text sample generation: Ten representative

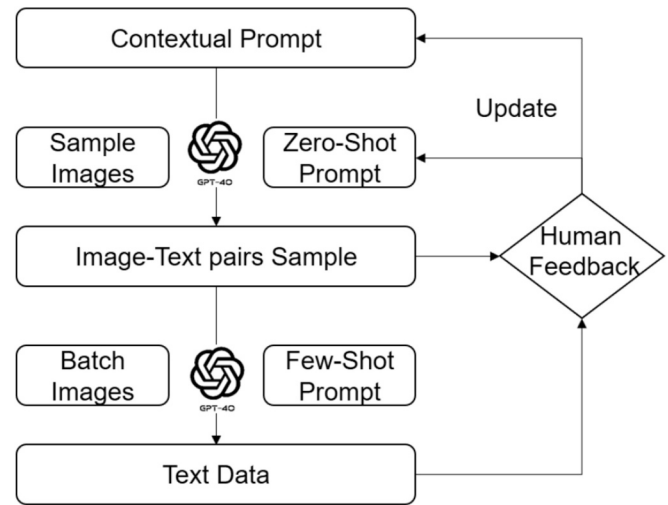


Fig. 2. Synthetic text data generation workflow.

images were selected, and textual descriptions were generated through iterative prompts refined by human feedback. Few-shot prompt batch processing: The generated image-text pairs were used to guide GPT-4o in processing the entire dataset. For large datasets, Python scripts and OpenAI's API (using GPT-4o) automated the prompt submission and text generation.

5.2.3. Topic modeling using BERTopic and evaluation

Topic modeling was employed to uncover latent spatial perception themes from the synthetic text data. BERTopic, a state-of-the-art topic modeling method, was used for its ability to generate coherent clusters through semantic vector analysis. Built on BERT (Bidirectional Encoder Representations from Transformers), BERTopic effectively embeds documents into high-dimensional semantic spaces, facilitating advanced clustering and topic representation (Devlin et al., 2019; Grootendorst, 2022).

The topic modeling workflow comprised the following steps (Fig. 3): Step 1, Embedding: The all-mpnet-base-v2 model was used to embed documents into a 768-dimensional vector space for semantic analysis. Step 2, Dimensionality Reduction: UMAP (Uniform Manifold Approximation and Projection) reduced the high-dimensional vectors while preserving both local and global data structures (McInnes et al., 2020). Step 3, Clustering: HDBSCAN effectively clustered spatial vectors, handling noise and identifying coherent topic groups with varying densities (Malzer & Baum, 2020). Step 4, Tokenization: CountVectorizer tokenized the text, with customized parameters for stop words and token length. Step 5, Weighting Scheme: c-TF-IDF was used to weight terms within each topic, highlighting their relative importance. Step 6, Representation: KeyBERTInspired optimized topic representations by aligning keywords with topic embeddings. After topic generation, results were manually reviewed to merge similar topics and discard irrelevant ones, ensuring coherence and relevance.

The model was evaluated using two widely accepted metrics: First, Topic coherence: Measured by normalized pointwise mutual information (NPMI), which scores associations between words within topics (Bouma, 2009). The model achieved an NPMI score of 0.178, indicating reasonable coherence. Second, Topic diversity: Measured as the percentage of unique words across all topics (Dieng et al., 2020). The model achieved a diversity score of 0.862, reflecting a broad and varied set of topics. While NPMI is better suited to classical topic models, its combination with modern metrics demonstrated the robustness of BERTopic for spatial perception analysis. Together, these metrics underscore the validity of the model and its ability to extract meaningful insights from the data.

Table 1
Spatial distribution and data classification.

Number	Section label	Spatial scope	Study subjects
Sec_1	Jingwei-St-W12-St	Jingwei Street to West 12th Street	Central Street
Sec_2	W12-St-W5-St	West 12th Street to West 5th Street	Central Street
Sec_3	W5-St-W2-St	West 5th Street to West 2nd Street	Central Street
Sec_4	W2-St-Youyi-Ro	West 2nd Street to Youyi Road	Central Street
Sec_5	Youyi-Ro-Flood-Con-Mon	Youyi Road to Flood Control Monument	Central Street
Sec_6	Flood-Con-Mon-Jiangshang-Cl	Flood Control Monument to Jiangshang Club	Stalin Park
Sec_7	Flood-Con-Mon-Binzhou-Rail-Bri	Flood Control Monument to Binzhou Railway Bridge	Stalin Park
Sec_8	Binzhou-Rail-Bri	Binzhou Railway Bridge	Stalin Park

Table 2
Contextual prompt engineering.

Domain knowledge	Spatial perception refers to the human ability to understand, recognize, and interact with space, encompassing the principles and mechanisms underlying spatial interactions. It serves as a fundamental basis for spatial planning, design, construction, and operation. ...
Contextual information	Topic modeling has emerged as a promising research method for studying spatial perception. Unlike traditional quantitative approaches with limited classifications, topic modeling enables a more accurate and systematic representation of spatial perception, capturing its complexity and multidimensionality more effectively. Harbin, the capital city of Heilongjiang Province in China, ... including examples of Art Nouveau, Eclecticism, early Modernism, and Chinese Baroque architecture. Harbin Central Street was established in 1898, ...offering a combination of shopping, cultural experiences, and architectural appreciation. Stalin Park is a significant greenbelt along the Songhua River, ... attracting a large number of both local residents and tourists every day.
Task instruction	Research Objective: To leverage GPT-4o's image-to-text technology for generating synthetic data and employ BERTopic for dynamic modeling to analyze global themes and local features of spatial perception. Input Data: The input consists of images captured by volunteers during on-site experiences, representing their spatial perception. Task Requirements: Based on the above domain knowledge, contextual data, and research objectives, use textual data to effectively represent the uploaded image dataset.
Input	
Output	At the entrance to Harbin Central Street stands a beautifully ornate wrought-iron archway, adorned with golden flower and leaf decorations, reflecting a strong European artistic style. ...the surrounding buildings exhibit typical European aesthetics, with yellow facades complementing the green trees, adding visual depth to the street. ... Lush greenery lines the center and sides of the street, creating an open and vibrant space. The area is bustling with pedestrians, primarily tourists, some of whom can be seen taking photos, reflecting the high tourism appeal and cultural significance of the district. In the background, floral arrangements and other landscaping elements add vitality and a touch of nature to the street.

6. Results

6.1. BERTopic topic modeling

A total of 39 spatial perception themes were identified through topic modeling. Using UMAP to embed these topics into a two-dimensional

space based on c-TF-IDF representations, a topic distance plot was created (Fig. 4). This visualization reveals potential clusters of topics, illustrating their underlying structure and relationships.

- 1) Cluster 1: Urban Facades and Architectural Heritage – Topics 0, 5, 10, 11, 16, and 38 focus on facades, building styles, and architectural details. This cluster underscores the significance of urban heritage by emphasizing intricate signage systems, street-level elements, and decorative aesthetics that define the built environment.
- 2) Cluster 2: Commercial Streets and Signage Systems – Topics 1, 17, 19, 24, and 35 capture the vibrancy of street-level commercial activity. They reflect the dynamic interplay of storefronts, signage, and shopping experiences, including distinctive cultural displays such as winter mascots, and cafes.
- 3) Cluster 3: Pedestrian and Recreational Spaces – Topics 2, 3, 9, 15, 18, 20, and 26 emphasize promenades, walkways, scenic views, and resting areas. This cluster illustrates how pedestrian-friendly infrastructure, such as riverside paths, benches, and fountains, supports social interactions and leisure activities.
- 4) Cluster 4: Natural Elements and Environmental Features – Topics 4, 6, 7, 13, 21, 28, 29, and 30 focus on natural aspects, including trees, greenery, unique pavement types like breadstone, and weather conditions. These elements are vital for urban livability, providing both aesthetic appeal and spaces for relaxation.
- 5) Cluster 5: Bridges and Riverside Elements – Topics 12, 14, 23, 25, 31, 32, 36, and 37 are associated with pedestrian bridges, riverside promenades, and scenic waterfront views. They highlight the importance of connectivity and the visual appeal of bridges and river scenes.
- 6) Cluster 6: Statues, Monuments, and Urban Art – Topics 22 and 27 concentrate on public art, including statues, sculptures, and monuments. These serve as cultural anchors, enhancing historical depth and decorative richness in urban spaces.
- 7) Cluster 7: Multicultural and Retail Spaces – Topics 8, 33, and 34 reflect the diverse commercial landscape, showcasing vibrant retail and dining experiences enriched by international cultural influences, such as Russian-themed stores and boutique signage.

Collectively, these clusters reveal the multifaceted nature of urban spatial perception in the Harbin Central Street District. The inter-topic distance map indicates that while some clusters form a cohesive core, others are more dispersed, reflecting localized nuances and distinct spatial identities. Overall, these findings highlight the importance of analyzing both global and local spatial perceptions to fully capture the complex interactions that shape urban experiences.

6.2. Extraction of global and local topics

Dynamic topic modeling incorporates spatial distribution data to differentiate between global and local themes. The frequency of each perception topic was calculated, and a threshold (≥ 1) was applied (Table 3). Topics that exceed the threshold in both Central Street and Stalin Park were classified as global, while those falling below it were categorized as local. To enhance accuracy, a refined spatial segmentation was implemented: Central Street was divided into five segments—Jinwei-St-W12-St, W12-St-W5-St, W5-St-W2-St, W2-St-Youyi-

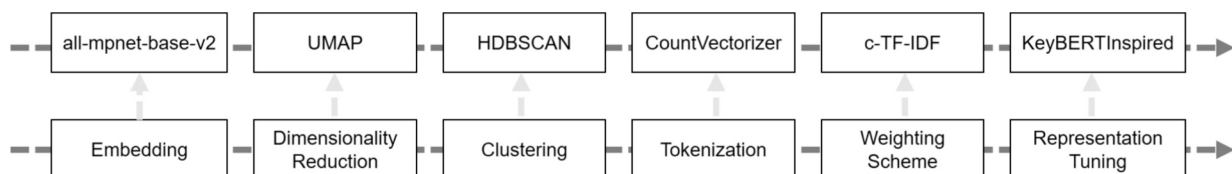


Fig. 3. Topic modeling workflow.

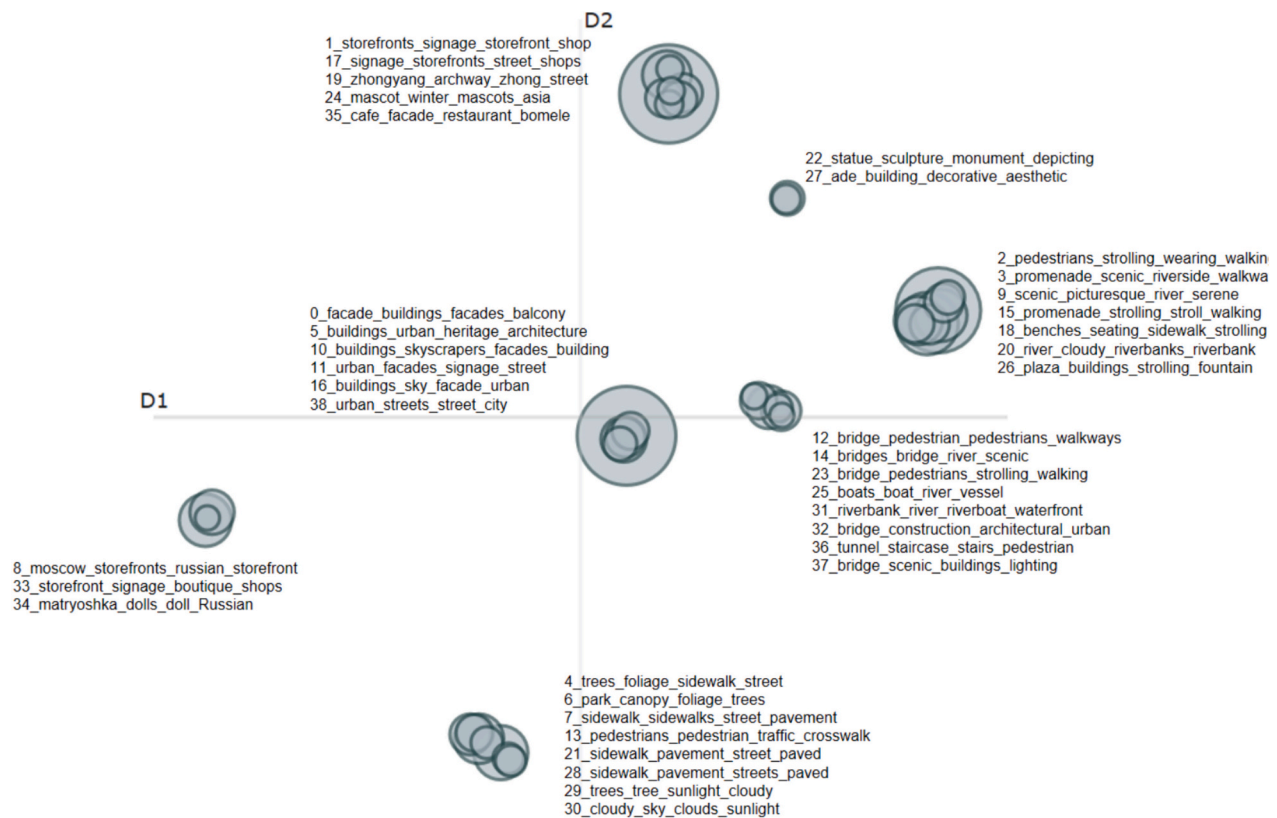


Fig. 4. Inter-topic distance map.

Ro, and Youyi-Ro-Flood-Cont-Mon—while Stalin Park was divided into three segments—Flood-Cont-Mon-Jiangshang-Club, Flood-Cont-Mon-Binzhou-Rail-Bri, and Binzhou-Rail-Bri. The analysis identified 15 global topics (e.g., topics 0, 1, 2, 4, 8, 10), 12 local topics unique to Central Street (e.g., topics 14, 15, 20), and 12 local topics specific to Stalin Park (e.g., topics 3, 6, 9) (Fig. 5). Overall, global topics comprised 38.4 % of the total, with local topics making up 30.8 % in both Central Street and Stalin Park.

6.3. Global patterns and local factors

The semantic proximity of topics in a high-dimensional vector space reflects their inherent similarities. To analyze spatial perception patterns, UMAP was utilized to project topics into a two-dimensional space. This visualization, created with the DataMapPlot library in Python, effectively illustrates topic weights and clustering distributions, offering an intuitive depiction of spatial perception themes.

6.3.1. Global patterns

The global topics are widely distributed and consistently present across both Central Street and Stalin Park, reflecting shared elements and common perceptions within the urban environment (Fig. 6). The clusters, represented by distinct color-coded regions, indicate the relative density and distribution of these topics.

1) Global Pattern 1: Pedestrian and Recreational Spaces

Pedestrian activity, seating areas, and strolling paths shape the dynamics of social interaction and urban livability. Key topics include 2, 13, 18, and 26. The emphasis on promenades, benches, and scenic routes highlights the importance of walkability and leisure in enhancing spatial experiences. These features promote physical activity, foster social encounters, and create a vibrant and inclusive urban atmosphere, contributing to the overall vitality of both Central Street and Stalin Park.

2) Global Pattern 2: Urban Facades and Architectural Heritage

Architectural elements, such as building facades and decorative details, play a pivotal role in defining the visual identity of urban environments. Key topics include 0, 10, and 27. The focus on historical structures and ornamental features underscores the aesthetic and cultural significance of architectural heritage. These elements not only preserve the district's historical essence but also enrich the sensory experience, fostering a visually cohesive and culturally immersive spatial environment.

3) Global Pattern 3: Commercial and Signage Systems

Commercial activities and storefront dynamics contribute significantly to the vibrancy of urban spaces. Key topics include 1, 8, and 35. Diverse shopping experiences, distinctive signage, and culturally themed displays infuse the streetfronts with dynamism and character. These commercial features enhance economic vitality and reinforce the district's identity as a dynamic and culturally diverse urban hub, catering to both residents and visitors.

4) Global Pattern 4: Statues, Monuments, and Public Art

Public art forms, including statues, sculptures, and monuments, enrich the cultural landscape and reinforce historical narratives. Key topic 22 highlights the symbolic importance of these elements in shaping spatial identity. These artistic and historical landmarks serve as visual anchors that deepen the spatial and cultural narrative of the district, offering tangible connections to its historical legacy.

5) Global Pattern 5: Natural and Environmental Features

Natural elements, such as trees, foliage, and atmospheric conditions, enhance the visual and ecological balance of urban environments. Key

Table 3
Spatial distribution frequency of perception topics.

Topic name	Central Street					Stalin Park		
	Jinwei-St- W12-St	W12-St- W5-St	W5-St- W2-St	W2-St- Youyi-Ro	Youyi-Ro- Flood-Cont- Mon	Flood-Cont-Mon- Jiangshang-Club	Flood-Cont-Mon- Binzhou-Rail-Bri	Binzhou- Rail-Bri
0_facade_buildings_facades_balcony	46	69	6	25	7	3	7	0
1_storefronts_signage_storefront_shop	55	61	22	11	3	0	7	2
2_pedestrians_strolling_wearing_walking	30	44	8	11	7	10	10	1
3_promenade_scenic_riverside_walkway	0	1	0	0	0	29	35	0
4_trees_foliage_sidewalk_street	10	25	5	7	0	1	1	0
5_buildings_urban_heritage_architecture	9	26	3	8	0	0	0	0
6_park_canopy_foliage_trees	0	0	0	0	0	22	24	0
7_sidewalk_sidewalks_street_pavement	15	17	5	4	0	0	0	0
8_moscow_storefronts_russian_storefront	25	6	5	1	0	0	3	0
9_scenic_picturesque_river_serene	0	0	0	0	0	10	29	0
10_buildings_skyscrapers_facades_building	4	10	3	9	1	3	3	2
11_urban_facades_signage_street	9	15	4	1	4	0	0	0
12_bridge_pedestrian_pedestrians_walkways	0	1	0	0	0	0	24	6
13_pedestrians_pedestrian_traffic_crosswalk	6	9	2	8	1	1	1	0
14_bridges_bridge_river_scenic	0	0	0	0	0	8	15	3
15_promenade_strolling_stroll_walking	0	0	0	0	0	10	16	0
16_buildings_sky_facade_urban	13	9	0	1	2	1	0	0
17_signage_storefronts_street_shops	10	8	2	1	4	0	0	0
18_benches_seating_sidewalk_strolling	7	9	4	1	0	1	3	0
19_zhongyang_archway_zhong_street	16	5	1	2	0	0	0	0
20_river_cloudy_riverbanks_riverbank	0	0	0	0	0	6	16	2
21_sidewalk_pavement_street_paved	8	12	2	0	0	1	0	0
22_statue_sculpture_monument_depicting	0	11	1	1	7	3	0	0
23_bridge_pedestrians_strolling_walking	0	0	0	1	0	0	18	3
24_mascot_winter_mascots_asia	11	7	1	0	0	0	2	0
25_boats_boat_river_vessel	0	0	0	0	0	8	11	0
26_plaza_buildings_strolling_fountain	1	3	0	0	3	7	5	0
27_ade_building_decorative_aesthetic	5	6	2	2	0	1	3	0
28_sidewalk_pavement_streets_paved	9	6	1	1	0	0	0	0
29_trees_tree_sunlight_cloudy	5	4	0	2	3	1	1	0
30_cloudy_sky_clouds_sunlight	0	1	1	1	2	3	7	1
31_riverbank_river_riverboat_waterfront	0	0	0	0	0	2	12	0
32_bridge_construction_architectural_urban	0	0	0	0	0	0	10	3
33_storefront_signage_boutique_shops	1	12	0	0	0	0	0	0
34_matryoshka_dolls_doll_russian	2	7	0	0	3	0	0	0
35_cafe_facade_restaurant_bomele	6	2	1	0	0	3	0	0
36_tunnel_staircase_stairs_pedestrian	0	0	0	11	0	0	0	0
37_bridge_scenic_buildings_lighting	0	0	0	0	0	0	9	1
38_urban_streets_street_city	5	3	2	0	0	0	0	0

topics include 4, 29, and 30. The integration of natural features into the urban fabric fosters harmony between built and natural spaces, creating a pleasant and sustainable urban atmosphere. These elements contribute to the aesthetic appeal and ecological resilience of both Central Street and Stalin Park.

Conclusion:

These global patterns reveal the intricate interplay of diverse urban elements—pedestrian activity, architectural heritage, commercial vibrancy, public art, and environmental features—that collectively shape a cohesive and dynamic spatial experience. The convergence of cultural, social, and environmental dimensions highlights the multifaceted nature of spatial perception in Central Street and Stalin Park. While some clusters exhibit strong thematic cohesion, others display more dispersed patterns, reflecting nuanced spatial identities and sensory experiences. These insights offer a robust framework for informing urban planning and design strategies that balance historical preservation, commercial vitality, and environmental sustainability, ultimately enhancing the spatial identity and livability of the district.

6.3.2. Local spatial perception factors on Central Street

The local topics identified in Central Street, as shown in the topics and documents map (Fig. 7), provide a nuanced understanding of the district’s spatial perception. The map, concentrated in the central and right sections, highlights five key factors that capture the area’s sensory and cultural richness.

1) Local Factor 1: Pedestrian Infrastructure and Urban Streetscape

Sidewalk infrastructure and vibrant streetscapes shape the pedestrian experience and enhance urban mobility in Central Street. Key topics include 7, 21, 28, 36, and 38. These topics emphasize the role of well-maintained walkways, diverse pavement types, and pedestrian-friendly designs that facilitate smooth movement and accessibility. By fostering social interaction and encouraging exploration, these features contribute to a dynamic and engaging urban environment that supports vibrant street life.

2) Local Factor 2: Architectural Identity and Urban Heritage

Building facades, intricate decorative details, and preserved historical structures define the architectural identity of Central Street. Key topics include 5, 11, and 16. These elements serve as visual markers that connect past and present, safeguarding the district’s cultural heritage while enhancing its aesthetic appeal. Maintaining the integrity of these architectural features ensures historical continuity and fosters a visually rich and culturally immersive urban landscape.

3) Local Factor 3: Commercial Signage and Boutique Storefronts

Distinctive signage systems and vibrant boutique storefronts define Central Street’s dynamic commercial atmosphere. Key topics include 17 and 33. These topics highlight the presence of culturally themed

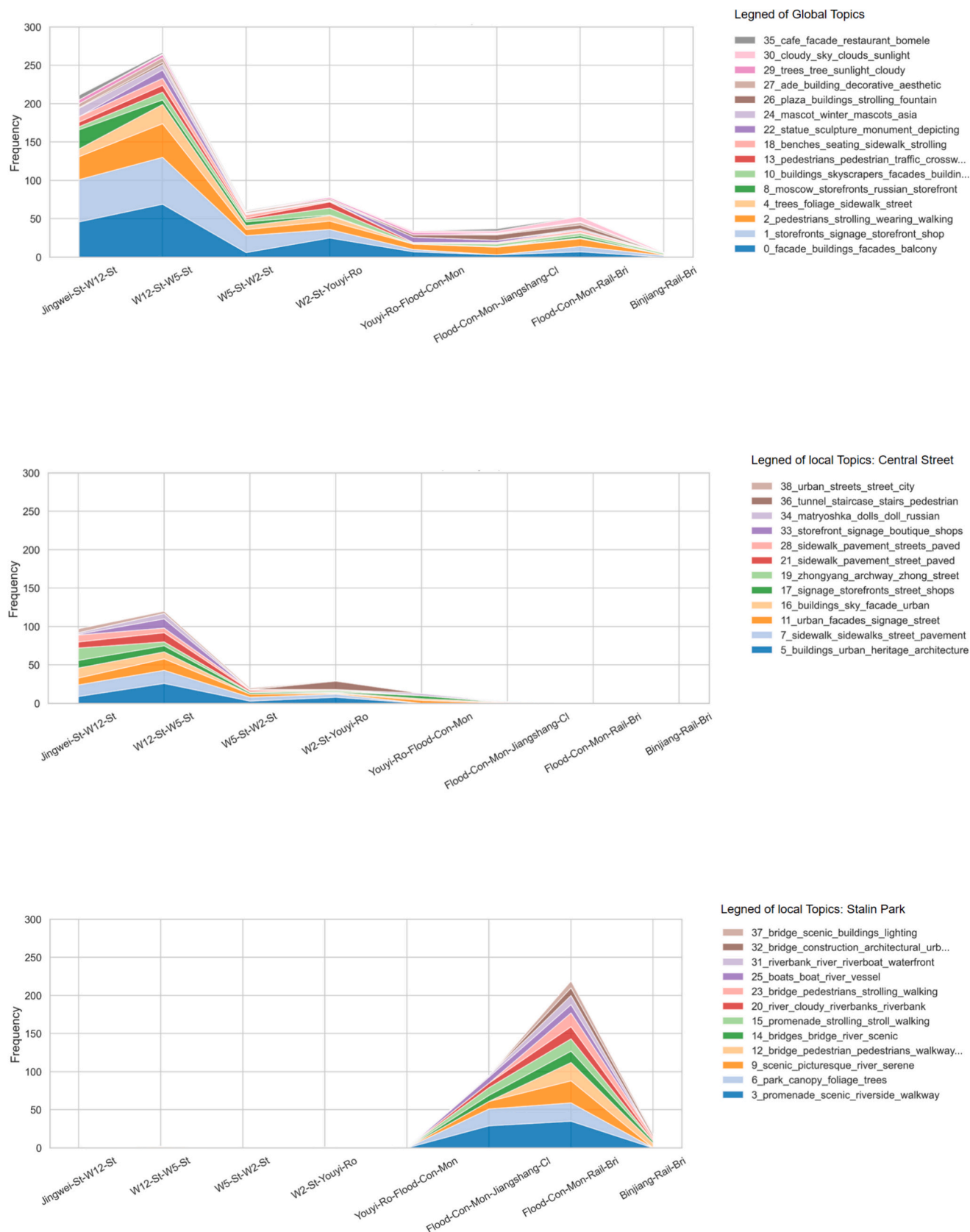


Fig. 5. Spatial distribution of topic frequencies.

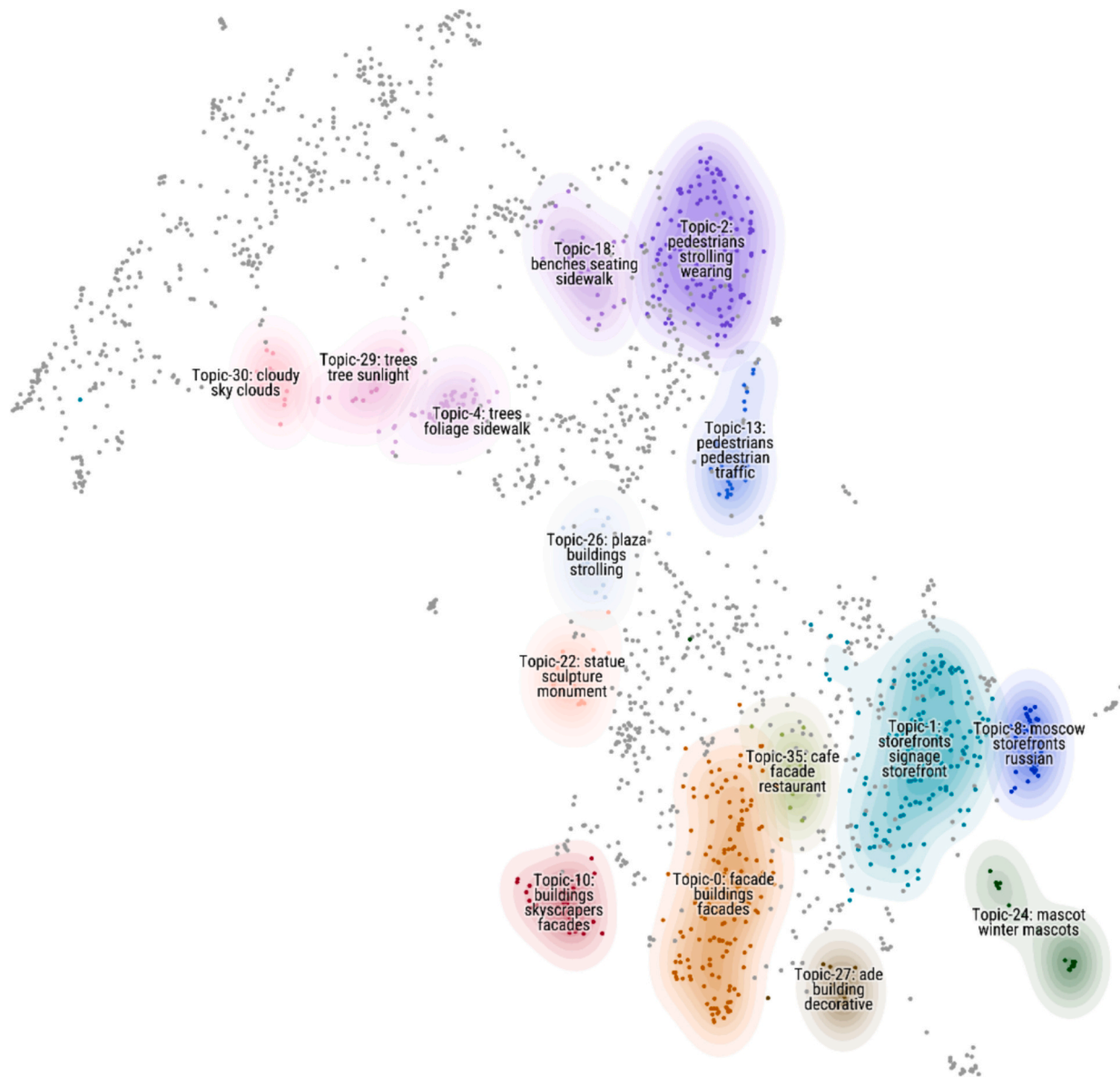


Fig. 6. Global topics and documents distribution

displays, specialized stores, and unique signage that create a visually engaging and diverse shopping environment. These elements not only enhance the street's commercial vitality but also reinforce its identity as a thriving cultural and economic hub.

4) Local Factor 4: Cultural and Decorative Influences

Cultural symbols and decorative motifs, such as matryoshka dolls, enrich Central Street's visual landscape by reflecting international cultural narratives. Key topic 34 captures the symbolic depth and visual diversity introduced by these elements, enhancing the district's multi-cultural identity. By integrating these symbols into the urban fabric, Central Street cultivates a rich and layered environment that reflects its cross-cultural influences and historical connections.

5) Local Factor 5: Iconic and Symbolic Landmarks

Iconic landmarks, such as Zhongyang Archway, serve as symbolic gateways that reinforce Central Street's historical and cultural identity. Key topic 19 underscores the importance of these landmarks as visual

anchors that establish a tangible link between past and present. Preserving these symbolic elements strengthens the district's spatial narrative and enhances its distinctive spatial perception.

Conclusion:

The local factors shaping Central Street underscore the intricate spatial dynamics that define the district's unique identity. From pedestrian-friendly infrastructure and architectural heritage to symbolic landmarks and commercial vibrancy, these elements collectively create a richly layered and multidimensional urban experience. Understanding these local factors provides a comprehensive framework for appreciating the complex spatial characteristics that define Central Street and inform future urban development strategies.

6.3.3. Local spatial perception factors in Stalin Park

The local topics document map for Stalin Park (Fig. 8) predominantly clusters in the upper-left section of the distribution, highlighting distinct spatial perception themes unique to this riverside park. Five key factors shape its identity:

1) Local Factor 1: Riverbanks and Scenic Views

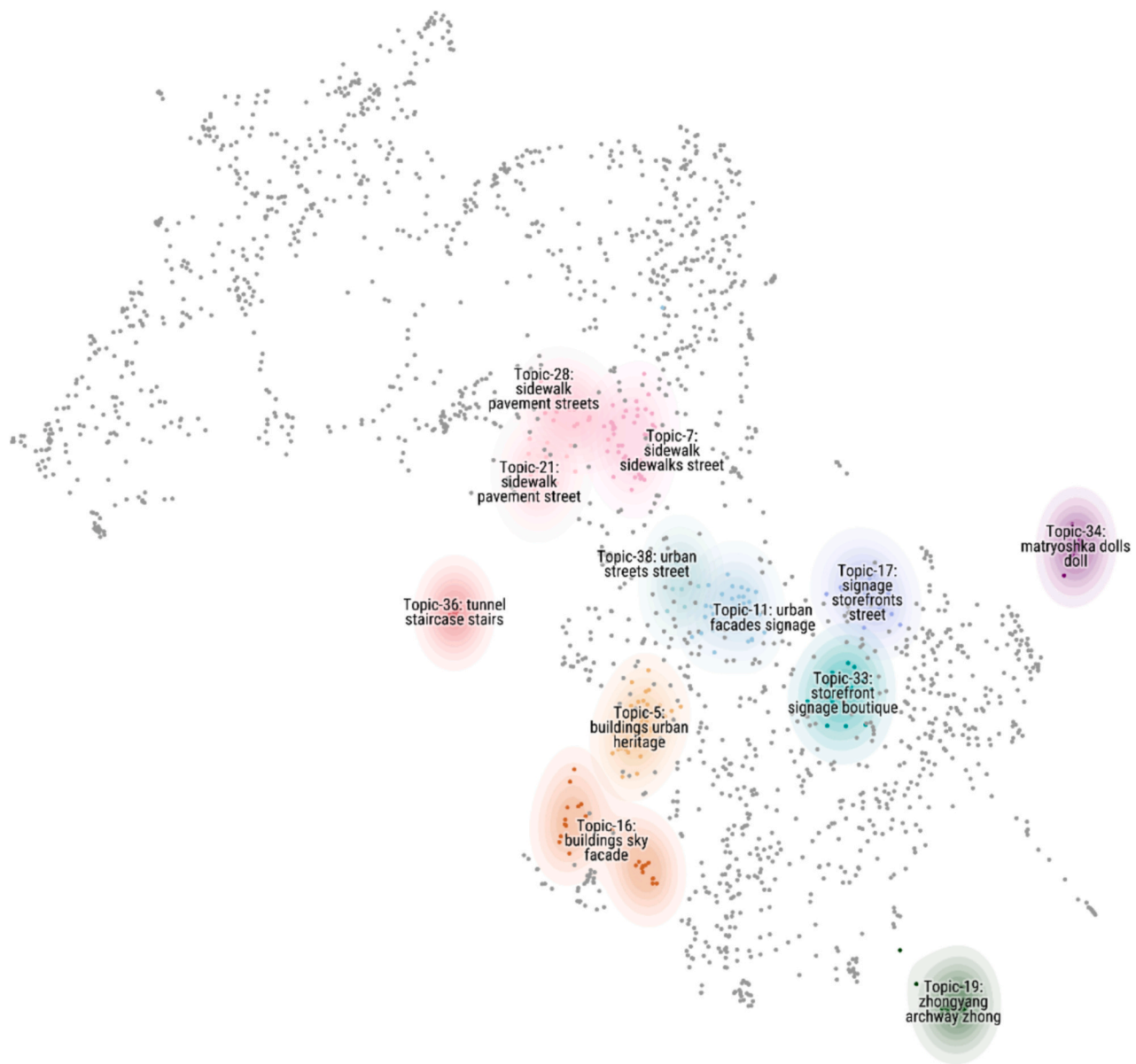


Fig. 7. Local topics and documents distribution of Central Street.

Scenic views along the riverbanks and picturesque landscapes shape the sensory perception of Stalin Park. Key topics include 9, 20, and 31. These topics highlight the tranquil ambiance created by riverside vistas, reflections on the water's surface, and interactions with riverboats. The visual richness of the riverfront fosters an immersive experience, establishing the park as a serene retreat amidst urban surroundings.

2) Local Factor 2: Bridges, Architectural Features, and Pedestrian Connectivity

Bridges and their architectural intricacies play a crucial role in enhancing both functional connectivity and visual appeal within Stalin Park. Key topics include 12, 14, 23, 32, and 37. Iconic structures such as the Binzhou Railway Bridge and the cableway crossing the river provide seamless pedestrian mobility while offering panoramic views that elevate the visitor experience. These architectural elements not only facilitate connectivity and link the park to Sun Island Park but also enrich the park's visual narrative, blending structural elegance with spatial functionality. Preserving these features reinforces the park's cultural identity and ensures a cohesive and accessible environment.

3) Local Factor 3: Promenades and Leisure Strolling

The park's promenades and scenic pathways cultivate a leisurely and calming atmosphere for visitors. Key topics include 3 and 15. These topics capture the relaxed pace of movement along picturesque routes, emphasizing the importance of promenades in promoting outdoor relaxation and social interaction. The integration of well-designed walkways and scenic landscapes creates an inviting environment that encourages recreational engagement.

4) Local Factor 4: Parks and Green Canopies

Lush greenery, dense foliage, and shaded canopies enhance the ecological balance and aesthetic appeal of Stalin Park. Key topic 6 highlights the significance of tree-lined walkways, natural foliage, and shaded areas in creating a cool and inviting environment. These natural features not only improve the sensory experience but also contribute to environmental sustainability, making Stalin Park a more comfortable and visually pleasing space.

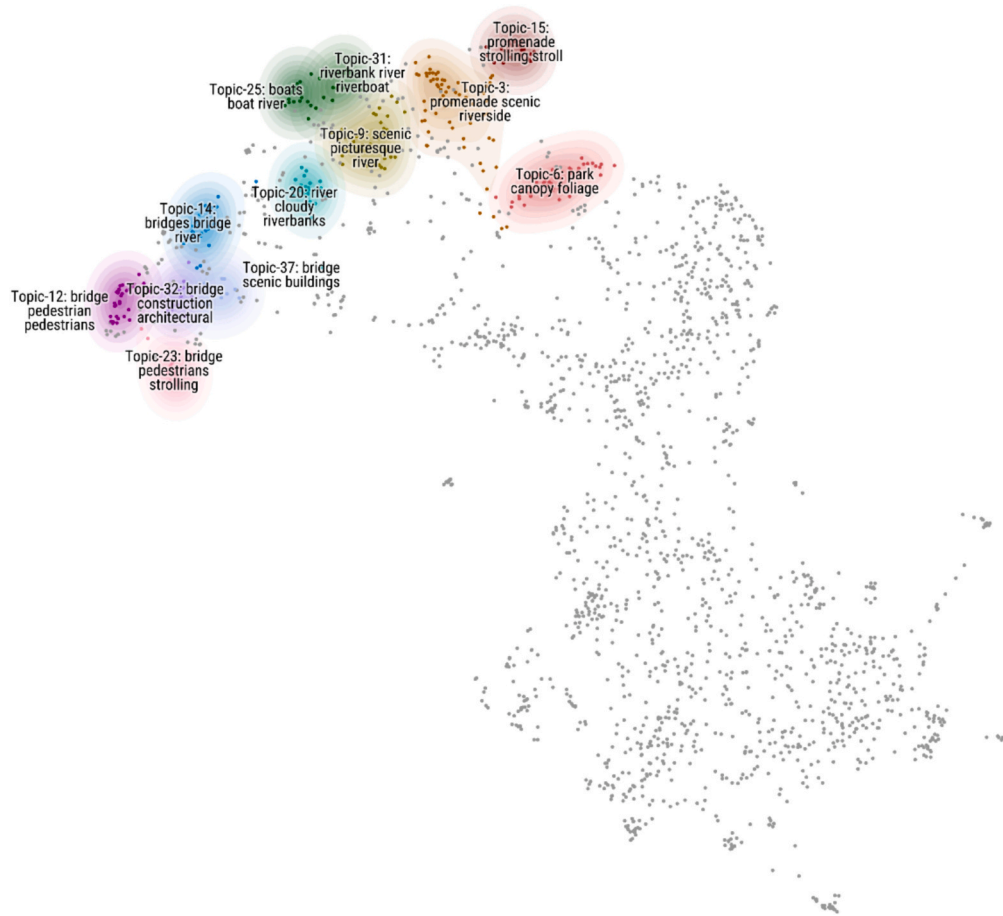


Fig. 8. Local topics and document distribution of Stalin part.

5) Local Factor 5: River Activities and Boat Dynamics

River activities, boat rides, and waterfront experiences inject vibrancy and interaction into the riverfront landscape. Key topics include 25 and 31. These topics emphasize the dynamic presence of riverboats and other water-based activities, offering opportunities for recreation and enriching the park's lively and engaging atmosphere. These activities contribute to the park's identity as an active and interactive public space, fostering a sense of connection between visitors and the natural environment.

Conclusion:

The localized factors shaping Stalin Park underscore the diverse spatial dynamics that define its unique identity. Scenic riverbank views, pedestrian-friendly pathways, shaded green canopies, and architectural landmarks collectively create a vibrant and multifaceted urban landscape. These elements contribute to the park's tranquil riverside ambience, accessible infrastructure, and dynamic recreational offerings. To preserve and enhance this identity, future efforts should prioritize maintaining natural landscapes, strengthening pedestrian connectivity, and diversifying waterfront activities. These strategies will ensure that Stalin Park remains an inclusive, vibrant, and culturally rich space that continues to enrich the lives of residents and visitors alike.

7. Discussion

This study achieved its objectives by leveraging advanced technologies, such as MLLMs and BERTopic, to develop a digital methodology for spatial perception research. A replicable and scalable workflow was designed and validated through an empirical case study of Harbin Central Street District. The analysis systematically identified global

patterns and local themes, capturing on-site spatial experiences and characteristics with accuracy. For instance, the historical architecture and dynamic commercial atmosphere of Central Street, alongside the natural environment and social interactions in Stalin Park, were effectively represented (Liu, 2003). Furthermore, the spatial perception themes identified in this study comprehensively cover physical spatial features, environmental quality, and individual experiences, aligning closely with established urban spatial metrics (Ewing et al., 2013). The discussion below highlights the methodological contributions of this research and its practical applications.

7.1. Digital representation and measurement of spatial perception

First, Representation Using Synthetic Text Data. This study introduces synthetic text data as a novel method for representing spatial perceptions, utilizing advanced MLLMs like GPT-4o to generate descriptive text from visual data. These models' capabilities in semantic understanding and spatial interpretation produce rich, detailed textual descriptions, advancing traditional approaches that rely on surveys, categorical data, or image segmentation. AI-generated text offers a scalable and precise representation of subjective spatial experiences, enabling a more nuanced understanding of the sensory and emotional dimensions of urban environments.

Second, High-Dimensional Topic Modeling. By employing BERT-based technologies such as BERTopic, this study utilizes high-dimensional semantic vector space modeling to uncover latent themes in spatial perception. Semantic clustering and c-TF-IDF modeling reveal systematic clusters of themes, from architectural elements to pedestrian activities. This approach surpasses traditional statistical methods, allowing for a deeper analysis of complex urban environments and the

relationships among key elements such as historical landmarks, commercial zones, and natural features.

Additionally, Dynamic Analysis of Global and Local Themes. Dynamic topic modeling further enhances the analysis by distinguishing between global patterns and localized factors in spatial perception. Integrating spatial distribution data into the topic modeling process captures both overarching themes and granular, site-specific details. Global themes provide insights into general spatial patterns, while local themes highlight unique characteristics, such as pedestrian pathways, riverside promenades, and cultural landmarks, that shape specific urban experiences.

7.2. Workflow and guidelines for spatial perception topic modeling

This study proposes a structured workflow for spatial perception research, integrating MLLMs and BERTopic modeling into a clear and adaptable framework (Fig. 9). The workflow consists of six key steps:

Step 1, Define Domain Knowledge and Core Themes. Refine spatial scales and research themes based on specific contexts, such as emotional analysis, post-occupancy evaluations, or cultural heritage studies.

Step 2, Generate Synthetic Text Data. Utilize image-to-text technologies, such as GPT-4o, to create descriptive textual data from visual inputs like photographs, street view imagery, and video frames. This synthetic text data forms the foundation for comprehensive spatial perception analysis.

Step 3, Conduct Topic Modeling. Apply BERTopic for semantic clustering and topic modeling, leveraging pre-trained transformer-based language models to generate contextually relevant spatial perception themes.

Step 4, Represent Topic Information. Generate outputs such as inter-topic distance maps to visualize relationships among themes and their spatial distributions, providing valuable insights into how perception patterns are organized.

Step 5, Extract Global and Local Topics. Distinguish between global topics that capture broad patterns and local topics that reflect detailed, context-specific perceptions. This dual-layered analysis enables a deeper exploration of spatial elements.

Step 6, Cluster and Visualize Topics. Use dimensionality reduction techniques like UMAP to create visualizations that depict the distribution of topics and documents across urban environments. These visualizations reveal underlying patterns and relationships within spatial perception data.

This workflow integrates MLLMs Fdriven text generation, semantic topic modeling, and advanced visualization techniques to provide a robust, scalable methodology for spatial perception research. It offers urban planners, designers, and policymakers actionable tools to analyze and represent how individuals experience and interact with urban environments.

7.3. Local practice and global applicability

Harbin Central Street District stands as a compelling case study to

validate the proposed workflow, demonstrating its effectiveness in dissecting spatial perceptions within a complex urban environment. Celebrated for its rich historical heritage and vibrant urban life, the district provides a unique lens to explore how individuals interact with historic architecture, bustling commercial areas, and adjacent natural landscapes.

The study's findings shed light on the intricate interplay among historical buildings, commercial dynamism, and public spaces in shaping urban experiences. It reveals that pedestrian flow and accessibility are pivotal in defining perceptions of vibrancy and usability, particularly evident in the contrasting yet complementary settings of Central Street's lively thoroughfares and Stalin Park's serene pathways. Well-preserved European-style architecture emerges as a cornerstone of cultural identity, enriching visitor experiences and emphasizing the role of historical continuity in spatial perception. The integration of green spaces—such as trees and riverbanks—further elevates perceptions of livability and ecological harmony, preserving the tranquil charm of Stalin Park. Commerce that resonates with the district's historical character sustains a unified yet dynamic atmosphere, while cultural elements like public art and landmarks deepen community ties and enhance the cultural tapestry. Together, these insights weave a nuanced narrative of how diverse urban components converge to influence spatial perception in the district.

Anchored in the unique context of Harbin, the workflow's methodology exhibits remarkable scalability and adaptability, positioning it as a versatile tool for analyzing historical districts and diverse urban environments worldwide (Li et al., 2025). Its global application enables cities to develop universal spatial perception models, deepening our understanding of how cultural heritage, historical legacies, and urban design converge to influence human interactions with urban spaces. This scalability not only affirms the workflow's practical value but also establishes it as a vital link between localized insights and broader urban theories, providing a robust framework to uncover the universal principles that shape varied urban experiences. By facilitating cross-cultural urban research, this approach equips planners and policymakers with the foresight to anticipate and address the perceptual dynamics that define cities across regions and cultures, fostering more informed and responsive urban strategies on a global scale.

7.4. Supporting urban planning and policy development

This study employs advanced tools—MLLMs and BERTopic modeling—to analyze spatial perceptions in Harbin Central Street District, generating actionable insights for urban planning and policy development. By uncovering global themes (e.g., pedestrian dynamics, architectural heritage, natural elements) and local nuances (e.g., Central Street, Stalin Park), it establishes a data-driven basis for designing inclusive, resilient, and human-centered urban environments.

First, Policy Implications. the study's findings inform a trio of interconnected policy shifts aimed at creating adaptive and inclusive urban spaces. These begin with human-centered design standards, mandating that all urban projects prioritize walkable pathways, heritage

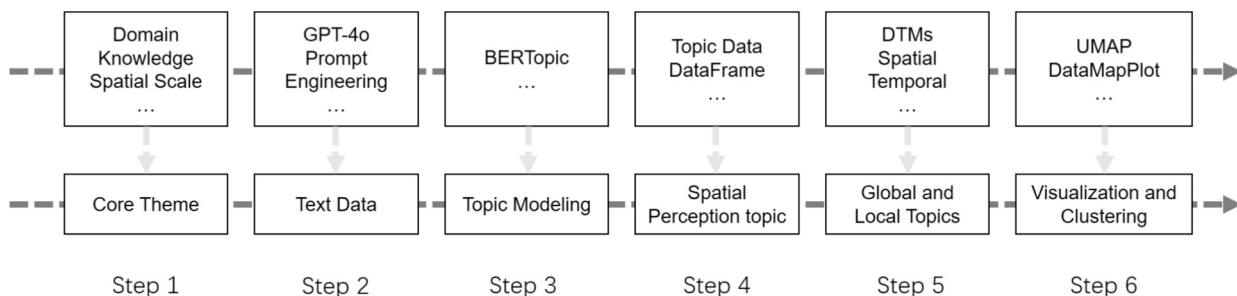


Fig. 9. Spatial perception analysis workflow.

preservation, and accessible green spaces, as seen in the preference for pedestrian-friendly zones on Central Street. This approach aligns development with human needs and cultural context, enhancing quality of life for residents and visitors. Building on this foundation, dynamic monitoring for real-time adaptation leverages the study's workflow to track spatial perceptions, enabling planners to assess intervention outcomes—like managing crowd flow during peak times—and adjust policies dynamically, ensuring strategies remain effective amid evolving conditions. Finally, community-informed planning integrates spatial perception data, such as calls for more cultural features, into decision-making, ensuring policies reflect local priorities and foster sustained relevance and support. Together, these shifts form a cohesive framework that balances human experiences with operational responsiveness, laying the groundwork for sustainable urban progress.

Second, Workflow Utilities. The proposed workflow—combining MLLMs driven text generation, BERTopic modeling, and dynamic topic analysis—offers a scalable toolkit for urban governance, structured as a logical sequence from insight generation to implementation. By continuously tracking shifts in spatial perceptions, such as seasonal overcrowding or event-driven changes, it empowers planners with real-time insights for proactive planning. Additionally, the workflow's ability to distinguish global themes (e.g., heritage protection) from local priorities (e.g., enhancing riverbank access in Stalin Park) supports targeted strategies for efficient resource use, optimizing resource allocation for maximum impact across diverse urban contexts. Furthermore, by integrating qualitative insights, like perceptions of natural elements, with quantitative data, such as traffic patterns and environmental metrics, it facilitates holistic decision-making through integrated data, ensuring urban designs are both functional and experientially enriching, meeting the dual demands of efficiency and livability. Together, these utilities enhance urban governance by aligning data-driven precision with human-centered outcomes, making cities more adaptive and livable.

In summary, this study showcases the power of MLLMs and topic modeling to transform spatial perception research into practical tools for urban planning and policymaking. By connecting data-driven insights to actionable strategies—such as improving pedestrian zones, preserving architectural heritage, and integrating natural elements—it fosters adaptive, inclusive urban landscapes. The proposed workflow equips planners with a dynamic, evidence-based framework to address evolving spatial needs, positioning Harbin's Central Street District—and cities globally—for sustainable, human-centered growth in the AI era.

8. Summary and future research

8.1. Summary

The rapid advancements in computational technology, particularly the emergence of MLLMs like GPT-4o, have ushered in a new era of algorithmic transformation, reshaping society and cities alike. This paradigm shifting calls for innovative theories and methodologies in spatial perception research to support urban planning and policymaking. At the same time, MLLMs present unprecedented opportunities for innovation in understanding urban environments. Responding to these dual imperatives, this study explores the integration of image-to-text technology and dynamic topic modeling as scalable and adaptive tools for analyzing urban spatial perception.

Using synthetic data and BERTopic modeling, the study identifies global patterns and local factors shaping spatial perception in two iconic areas of Harbin: Central Street and Stalin Park. Central Street is defined by its vibrant commercial activity, historical architecture, and lively storefronts, encapsulating the dynamic and bustling character of this historic district. In contrast, Stalin Park emphasizes natural elements such as riverbanks, promenades, and tree-lined walkways, presenting a serene space for relaxation and scenic enjoyment.

This research establishes a replicable workflow for spatial perception

analysis and provides actionable insights for urban planning and policymaking. The findings demonstrate the potential of cutting-edge LLMs technologies to offer systematic information support for urban decision-making, such as dynamic monitoring platforms or integrations with more complex urban workflows. The adaptability of this approach allows for application in other urban areas or public spaces globally, positioning it as a valuable tool in the field of computational urban science.

8.2. Limitations and future research

While this study makes significant contributions to spatial perception research, several limitations must be addressed:

First, urban spatial perception is inherently multisensory, yet this study predominantly relies on visual data. To enrich future research, integrating additional sensory modalities—such as audio-to-text conversion or crowdsourced social media data—could offer a more holistic view of the sensory experience. For example, capturing soundscapes, ambient noise, or even tactile impressions could unveil fresh perspectives on spatial perception, as supported by prior work (Bruce et al., 2015; Thibaud, 2011). Broadening the sensory lens would yield deeper insights into how diverse sensory inputs shape perceptions of urban environments.

Additionally, this study's reliance on static spatial images constrains its capacity to examine real-time shifts in spatial perception. Future research could address this by adopting dynamic analysis, leveraging live image feeds, video data, or real-time social media updates. Such an approach would allow exploration of temporal fluctuations in perception—spanning day-to-night transitions, seasonal changes, or the effects of special events—better reflecting the fluid, evolving nature of urban experiences.

Furthermore, focusing exclusively on Harbin limits the generalizability of this study's findings. Future research should expand to include multiple cities and cultural contexts to identify universal principles and location-specific factors influencing spatial perception. Comparative studies across diverse urban environments could reveal both global trends and unique local characteristics, enriching our understanding of how spatial perceptions vary by culture, geography, and historical context.

Last, as LLMs and other AI technologies continue to advance, their semantic understanding and spatial perception capabilities will significantly improve. These advancements are likely to reshape how urban spaces are perceived and experienced. Future research should examine the evolving relationship between AI-driven technologies and spatial perception patterns, exploring whether existing findings remain consistent as AI increasingly influences urban environments.

Addressing these limitations will enhance theoretical and practical insights into spatial perception while fostering the development of more advanced AI-driven tools for urban analysis. These tools can pave the way for creating emotionally responsive urban environments that adapt to the evolving needs and preferences of residents and visitors, thereby advancing both the theoretical and applied dimensions of urban spatial perception in the AI era.

CRedit authorship contribution statement

Haixuan Zhu: Writing – review & editing, Writing – original draft, Visualization, Software, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Jiang Chang:** Resources, Investigation, Data curation. **Xinyu An:** Writing – review & editing, Investigation, Data curation. **Shilin Li:** Investigation, Data curation.

Declaration of competing interest

The authors declare the following financial interests/personal

relationships which may be considered as potential competing interests:

Haixuan Zhu reports financial support was provided by Department of Science and Technology of Heilongjiang Province, China. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors express their sincere gratitude to Prof. Pengjun Zhao, Prof. Simon Joss, and the anonymous reviewers for their invaluable feedback on earlier versions of this article. This study was conducted as part of the ongoing research on computational city science theories and methodologies. It received financial support from the Department of Science and Technology of Heilongjiang Province, China (Grant No. G2023062).

Data availability

This study represents a stage of guiding students' thesis research and serves as an exploration of a specific case study and workflow. Upon completion of subsequent research, relevant data and materials will be made publicly available on GitHub.

References

- Biljecki, F., & Ito, K. (2021). Street view imagery in urban analytics and GIS: A review. *Landscape and Urban Planning*, 215, Article 104217. <https://doi.org/10.1016/j.landurbplan.2021.104217>
- Blei, D. M., & Lafferty, J. D. (2006, June). Dynamic topic models. In *Proceedings of the 23rd international conference on machine learning* (pp. 113–120).
- Blei, D. M., Ng, A. Y., & Jordan, M. I. (2003). Latent dirichlet allocation. *Journal of Machine Learning Research*, 3(Jan), 993–1022.
- Böcker, L., Dijst, M., Faber, J., & Hellich, M. (2015). En-route weather and place valuations for different transport mode users. *Journal of Transport Geography*, 47, 128–138. <https://doi.org/10.1016/j.jtrangeo.2015.06.003>
- Bouma, G. J. (2009). Normalized (pointwise) mutual information in collocation extraction. *Proceedings of GSCL*, 30, 31–40.
- Bruce, N., Condie, J., Henshaw, V., & Payne, S. R. (2015). Analysing olfactory and auditory senses in English cities: Sensory expectation and urban environmental perception. *Ambiances. Environnement sensible, architecture et espace urbain*.
- Chen, B., Xu, Z., Kirmani, S., Ichter, B., Sadigh, D., Guibas, L., et al. (2024). Spatialvlm: Endowing vision-language models with spatial reasoning capabilities. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 14455–14465).
- Chowdhary, K. R. (2020). Natural language processing. In *Fundamentals of artificial intelligence* (pp. 603–649). New Delhi: Springer. https://doi.org/10.1007/978-81-322-3972-7_19
- Cresswell, T. (1992). The crucial 'where' of graffiti: A geographical analysis of reactions to graffiti in New York. *Environment and Planning D: Society and Space*, 10(3), 329–344. <https://doi.org/10.1068/d100329>
- De Jong, M., Joss, S., & Taeihagh, A. (2024). Smart cities as spatial manifestations of 21st century capitalism. *Technological Forecasting and Social Change*, 202, Article 123299. <https://doi.org/10.1016/j.techfore.2024.123299>
- Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. arXiv. <https://doi.org/10.48550/arXiv.1810.04805>
- Dieng, A. B., Ruiz, F. J. R., & Blei, D. M. (2020). Topic modeling in embedding spaces. *Transactions of the Association for Computational Linguistics*, 8, 439–453.
- Doppelt, G. (2007). Reconstructing scientific realism to rebut the pessimistic meta-induction. *Philosophy of Science*, 74(1), 96–118.
- Downs, R. M., & Stea, D. (Eds.). (1973). *Image & environment: Cognitive mapping and spatial behavior*. AldineTransaction.
- Dubey, A., Naik, N., Parikh, D., Raskar, R., & Hidalgo, C. A. (2016). *Deep learning the city: Quantifying urban perception at a global scale*. ArXiv160801769 Cs.
- Dunning, T. (2012). *Natural experiments in the social science: A design-based approach*. Cambridge University Press.
- Eliasson, I., Knez, I., Westerberg, U., Thorsson, S., & Lindberg, F. (2007). Climate and behaviour in a Nordic city. *Landscape and Urban Planning*, 82(1–2), 72–84. <https://doi.org/10.1016/j.landurbplan.2007.01.020>
- Ewing, R., Clemente, O., Neckerman, K. M., Purciel-Hill, M., Quinn, J. W., & Rundle, A. (2013). *Measuring urban design: Metrics for livable places* (Vol. 200). Springer.
- Gabriel, I. (2020). Artificial intelligence, values and alignment. *Minds and Machines*, 30(3), 411–437. <https://doi.org/10.1007/s11023-020-09539-2>
- Garling, T. (1972). Studies in visual perception of architectural spaces and rooms: V. Aesthetic preferences. *Scandinavian Journal of Psychology*, 13. <https://doi.org/10.1111/j.1467-9450.1972.tb00070.x>
- Gehl, J. (1987). *Life between buildings: Using public space*. translated by Jo Koch, Van Nostrand Reinhold, New York. (ISBN 978-87-7407-360-4).
- Gibson, J. J. (1950). *The perception of the visual world*. Houghton Mifflin.
- Gonzalez, D., Rueda-Plata, D., Acevedo, A. B., Duque, J. C., Ramos-Poll'an, R., Betancourt, A., & García, S. (2020). Automatic detection of building typology using deep learning methods on street level images. *Building and Environment*, 177, Article 106805. <https://doi.org/10.1016/j.buildenv.2020.106805>
- Grootendorst, M. (2022). BERTopic: Neural topic modeling with a class-based TF-IDF procedure. arXiv. <https://doi.org/10.48550/arXiv.2203.05794>
- Hardin, C. L., & Rosenberg, A. (1982). In defense of convergent realism. *Philosophy of Science*, 49(4), 604–615.
- Heft, H. (2013). Environment, cognition, and culture: Reconsidering the cognitive map. *Journal of Environmental Psychology*, 33, 14–25.
- Huh, M., Cheung, B., Wang, T., & Isola, P. *The platonic representation hypothesis*. (). arXiv preprint arXiv:2405.07987.
- Ito, K., Kang, Y., Zhang, Y., Zhang, F., & Biljecki, F. (2024). Understanding urban perception with visual data: A systematic review. *Cities*, 152, Article 105169. <https://doi.org/10.1016/j.cities.2024.105169>
- Kaplan, R. (1973). Some psychological benefits of gardening. *Environment and Behavior*, 5(2), 145–162. <https://doi.org/10.1177/001391657300500202>
- Khurana, D., Koli, A., Khatter, K., & Singh, S. (2023). Natural language processing: State of the art, current trends and challenges. *Multimedia Tools and Applications*, 82(3), 3713–3744. <https://doi.org/10.1007/s11042-022-13428-4>
- Kitchin, R., & Blades, M. (2002). *The cognition of geographic space* (Vol. 4). London: Ib Tauris.
- Klemm, W., Heusinkveld, B. G., Lenzholzer, S., & van Hove, B. (2015). Street greenery and its physical and psychological impact on thermal comfort. *Landscape and Urban Planning*, 138, 87–98. <https://doi.org/10.1016/j.landurbplan.2015.02.009>
- Kosinski, M. (2024). Evaluating large language models in theory of mind tasks. *Proceedings of the National Academy of Sciences*, 121(45), Article e2405460121. <https://doi.org/10.1073/pnas.2405460121>
- Lenzholzer, S., & Koh, J. (2010). Immersed in microclimatic space: Microclimate experience and perception of spatial configurations in Dutch squares. *Landscape and Urban Planning*, 95(1–2), 1–15. <https://doi.org/10.1016/j.landurbplan.2009.10.013>
- Li, F. (2024). *With spatial intelligence, AI will understand the real*. TED: World. <https://www.youtube.com/watch?v=y8NtMz7VGmU>
- Li, P., Zhang, H., Li, W., Huang, D., Guo, Z., Chen, J., Zhang, J., Song, X., Zhao, P., Yan, J., Ryosuke, S., & Koshizuka, N. (2025). GeoAvatar: A big mobile phone positioning data-driven method for individualized pseudo personal mobility data generation. *Computers, Environment and Urban Systems*, 119, Article 102252. <https://doi.org/10.1016/j.compenvurbsys.2025.102252>
- Liddy, E. D. (2001). Natural language processing. In *Encyclopedia of library and information science* (2nd ed., pp. 1–16). NY: Marcel Decker, Inc.
- Liu, B. (2012). Sentiment analysis and opinion mining. *Synthesis Lectures on Human Language Technologies*, 5(1), 1–167.
- Liu, S. (2003). *Harbin's urban architectural transformation and model exploration: 1898–1949*. China Architecture & Building Press. ISBN: 711205995X, 9787112059959.
- Lynch, K. (1960). *The image of the City*. MIT Press.
- Ma, X., Ma, C., Wu, C., Xi, Y., Yang, R., Peng, N., Zhang, C., & Ren, F. (2021). Measuring human perceptions of streetscapes to better inform urban renewal: A perspective of scene semantic parsing. *Cities*, 110, Article 103086. <https://doi.org/10.1016/j.cities.2020.103086>
- Malekzadeh, M., Willberg, E., Torkko, J., & Toivonen, T. (2025). Urban attractiveness according to ChatGPT: Contrasting AI and human insights. *Computers, Environment and Urban Systems*, 117, Article 102243. <https://doi.org/10.1016/j.compenvurbsys.2024.102243>
- Malzer, C., & Baum, M. (2020, September). A hybrid approach to hierarchical density-based cluster selection. In *2020 IEEE international conference on multisensor fusion and integration for intelligent systems (MFI)* (pp. 223–228). IEEE.
- McGinn, A. P., Evenson, K. R., Herring, A. H., Huston, S. L., & Rodriguez, D. A. (2007). Exploring associations between physical activity and perceived and objective measures of the built environment. *Journal of Urban Health*, 84, 162–184. <https://doi.org/10.1007/s11524-006-9136-4>
- McInnes, L., Healy, J., & Melville, J. (2020). UMAP: Uniform manifold approximation and projection for dimension reduction. arXiv. <https://doi.org/10.48550/arXiv.1802.03426>
- Mikolov, T., Chen, K., Corrado, G., & Dean, J. (2013). Efficient estimation of word representations in vector space. arXiv preprint arXiv:1301.3781.
- Montello, D., Goodchild, M., Gottsegen, J., & Fohl, P. (2003). Where's downtown? Behavioral methods for determining referents of vague spatial queries. *Spatial Cognition & Computation*, 3(2–3), 185–204. <https://doi.org/10.1080/13875868.2003.9683761>
- Nagata, S., Nakaya, T., Hanibuchi, T., Amagasa, S., Kikuchi, H., & Inoue, S. (2020). Objective scoring of streetscape walkability related to leisure walking: Statistical modeling approach with semantic segmentation of Google Street View images. *Health & Place*, 66, Article 102428. <https://doi.org/10.1016/j.healthplace.2020.102428>
- Newton-Smith, W. (1981). The rationality of science. In *International library of philosophy, psychology, and scientific method*. Routledge & Kegan Paul. ISBN 9780710009135.
- Putnam, H. (1982). Three kinds of scientific realism. *The Philosophical Quarterly* (1950), 32(128), 195–200.
- Salloum, S. A., Khan, R., & Shaalan, K. (2020). A survey of semantic analysis approaches. In *Proceedings of the international conference on artificial intelligence and computer vision (AICV2020)*, 61–70. Springer International Publishing.

- Sperling, J. (2024). How AI is fueling a 'cognitive industrial revolution. In *Digital future: Insights, artificial intelligence (AI), marketplace design*. Published April 5, 2024. Retrieved from <https://business.columbia.edu/insights/digital-future/how-ai-fuelin-g-cognitive-industrial-revolution>.
- Thibaud, J. P. (2011). The sensory fabric of urban ambiances. *The Senses and Society*, 6 (2), 203–215.
- Touvron, H., Lavril, T., Izacard, G., Martinet, X., Lachaux, M.-A., Lacroix, T., ... Lample, G. (2023). *LLaMA: Open and efficient foundation language models*. arXiv. <https://doi.org/10.48550/arXiv.2302.13971>.
- Tuan, Y. F. (1977). *Space and place: The perspective of experience*. University of Minnesota Press.
- Wang, L., Han, X., He, J., & Jung, T. (2022). Measuring residents' perceptions of city streets to inform better street planning through deep learning and space syntax. *ISPRS Journal of Photogrammetry and Remote Sensing*, 190, 215–230. <https://doi.org/10.1016/j.isprsjprs.2022.06.011>
- Wang, R., Liu, Y., Lu, Y., Zhang, J., Liu, P., Yao, Y., & Grekousis, G. (2019). Perceptions of built environment and health outcomes for older Chinese in Beijing: A big data approach with street view images and deep learning technique. *Computers, Environment and Urban Systems*, 78, Article 101386. <https://doi.org/10.1016/j.compenvurbsys.2019.101386>
- Wankhade, M., Rao, A. C. S., & Kulkarni, C. (2022). A survey on sentiment analysis methods, applications, and challenges. *Artificial Intelligence Review*, 55(7), 5731–5780.
- Wu, J., Feng, Z., Peng, Y., Liu, Q., & He, Q. (2019). Neglected green street landscapes: A reevaluation method of green justice. *Urban Forestry & Urban Greening*, 41, 344–353. <https://doi.org/10.1016/j.ufug.2019.05.004>
- Xi, Z., Chen, W., Guo, X., He, W., Ding, Y., Hong, B., ... Gui, T. (2023). *The rise and potential of large language model based agents: A survey*. arXiv preprint. <https://doi.org/10.48550/arXiv.2309.07864>.
- Zhu, H., Wu, C., & Zhao, Z. (2024). Research on function combination model of historical blocks based on access sequence: A case study of Harbin central street[J]. *Journal of Urban and Regional Planning*, 16(1), 117–134.